

Quiz (Habanero)

- Q1.** The average temperature in a city over 5 days was 25°C . If the temperatures were 22°C , 24°C , 26°C , 28°C , and x , what is the value of x ?
- (A) 20°C
(B) 22°C
(C) 24°C
(D) 26°C
(E) 30°C
- Q2.** A researcher recorded the amount of rainfall in a region over 7 days: 12mm , 15mm , 18mm , 12mm , 15mm , 20mm , x . The mode of the data set is 15mm . What is the value of x ?
- (A) 10mm
(B) 12mm
(C) 15mm
(D) 18mm
(E) 20mm
- Q3.** The median of the data set 2, 4, 6, 8, 10 is)
- (A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q4.** A city's temperature readings for a week were 10°C , 12°C , 15°C , 18°C , 20°C , 22°C , 25°C . What is the median temperature?
- (A) 15°C
(B) 16°C
(C) 17°C
(D) 18°C
(E) 20°C
- Q5.** The average daily rainfall in a region over 10 days was 20mm . If it rained 15mm on the 1st day, 25mm on the 2nd day, 30mm on the 3rd day, 20mm on the 4th day, 15mm on the 5th day, 10mm on the 6th day, 25mm on the 7th day, 30mm on the 8th day, and x on the 9th and 10th days, what is the total rainfall on the 9th and 10th days?
- (A) 10mm
(B) 20mm
(C) 30mm
(D) 40mm
(E) 50mm
- Q6.** In a data set, the mean is 10, the median is 12, and the mode is 15. Which of the following statements is true?
- (A) The data set is skewed to the left
(B) The data set is skewed to the right
(C) The data set is symmetric
(D) The mean is equal to the median
(E) The mode is equal to the mean
- Q7.** The amount of carbon dioxide emissions in a city over 5 years were 1000, 1200, 1500, 1800, 2000. The median of the data set is
- (A) 1200
(B) 1300
(C) 1400
(D) 1500
(E) 1600
- Q8.** A researcher recorded the pH levels of a lake over 6 days: 6.5, 7.0, 7.5, 8.0, 8.5, x . The mean pH level is 7.5. What is the value of x ?
- (A) 6.0
(B) 6.5
(C) 7.0
(D) 8.0
(E) 9.0

Open Ended Questions (FRQ)

- Q9.** The average temperature in a city over the summer months was 28°C . If the temperature readings for each month were 25°C , 30°C , x , what is the value of x ? Show your work and explain your reasoning.
- Q10.** A data set has the following values: 2, 4, 6, 8, 10, 12, x . Find the median and mode of the data set if $x = 8$. Explain your reasoning and show your work.

Chef's Tips

- Read the questions carefully before answering
- Use a calculator if necessary
- Show your work for free response questions